

Data Sources and Licenses

NeuroSleep Data Sources

This document describes all datasets used in the NeuroSleep project, their access requirements, and licensing terms.

1. Sleep-EDF Expanded (Primary Dataset)

Property	Value
Provider	PhysioNet / MIT Laboratory for Computational Physiology
URL	https://www.physionet.org/content/sleep-edfx/1.0.0/ (https://www.physionet.org/content/sleep-edfx/1.0.0/).
DOI	10.13026/C2X676
Access	Open Access (no registration required)
License	Open Data Commons Attribution License v1.0
Subjects	197 whole-night recordings
Channels	EEG (Fpz-Cz, Pz-Oz), EOG (horizontal), EMG (submental)
Sampling Rate	100 Hz (EEG/EOG), 1 Hz (EMG envelope)
Scoring	Rechtschaffen & Kales (R&K) manual scoring
Epoch Duration	30 seconds

Subset Used

Our primary experiments use **153 healthy controls** (SC4xxxx subjects) from the Sleep Cassette study, recorded 1987-1991.

Note: Subject SC4013 was lost due to a failing cassette (per PhysioNet documentation).

Citation

B Kemp, AH Zwinderman, B Tuk, HAC Kamphuisen, JLL Obery . Analysis of a sleep-dependent neuronal feedback loop: the slow-wave microcontinuity of the EEG. IEEE-BME 47(9):1185-1194 (2000).

2. SHHS (Planned External Dataset)

Property	Value
Provider	National Sleep Research Resource (NSRR)
URL	https://sleepdata.org/datasets/shhs/ (https://sleepdata.org/datasets/shhs/).

Property	Value
Access	Data Use Agreement required — must register at sleepdata.org
License	NSRR Data Use Agreement
Subjects	SHHS Visit 1: 5,793 subjects; Visit 2: 2,651 subjects
Channels	Variable montage (must verify from documentation)
Scoring	AASM-compatible staging
Format	EDF (PSG) + XML (staging annotations)

Access Requirements

1. Register at <https://sleepdata.org> (<https://sleepdata.org>).
2. Complete Data Use Agreement
3. Download files via NSRR portal or AWS CLI
4. **Do not redistribute raw data**

Usage Plan

SHHS will be used for:

1. **N1 enrichment** — controlled injection of SHHS N1 samples into Sleep-EDF training
2. **Cross-dataset pretraining** — train on SHHS + Sleep-EDF, evaluate on Sleep-EDF
3. **Domain adaptation** — LoRA adaptation from SHHS-pretrained base to Sleep-EDF target

Citation

Quan SF, Howard BV, Iber C, et al. The Sleep Heart Health Study: a study of obstructive sleep apnea and cardiovascular disease in a community sample. Am J Respir Crit Care Med. 1997;155(3):1070-1077.

3. Data Processing Provenance

All processed data includes provenance tracking:

```
dataset: sleep_edf_expanded
version: 1.0.0
preprocessing:
  sampling_rate: 100 Hz
  epoch_seconds: 30
  bandpass: [0.5, 35.0] Hz
  notch: 50 Hz
  normalization: z-score
scoring_mapping: R&K → AASM canonical
  W → Wake (0)
  1 → N1 (1)
  2 → N2 (2)
  3,4 → N3 (3)
  R → REM (4)
  M,? → excluded
```

4. Redistribution Rules

Dataset	Raw Data	Processed Cache	Model Checkpoints
Sleep-EDF	☐ Allowed (ODC-BY)	☐ Allowed	☐ Allowed
SHHS	☐ Requires NSRR permission	☐ Check DUA	☐ Allowed
Our Models	N/A	☐ Allowed	☐ Allowed

5. File Organization

```

data/
├─ raw/
|   └─ sleep_edf/           # Raw EDF files (from PhysioNet)
|       └─ SC4001E0-PSG.edf # 16 PSG files downloaded
|       └─ SC4002E0-PSG.edf
|       └─ ...
|       └─ *-Hypnogram.edf  # Hypnogram annotations
├─ cache/
|   └─ sleep_edf/           # Processed NPZ files (15 subjects)
|       └─ SC4001_night0.npz
|       └─ SC4002_night0.npz
|       └─ ...
|       └─ checksums.json
|   └─ shhs/               # Processed NPZ files (SHHS, future)
├─ manifests/
|   └─ sleep_edf.csv        # Original 4-subject manifest
|   └─ sleep_edf_expanded.json # 15-subject manifest

```

Current State

- **16 PSG files** downloaded from PhysioNet (Sleep-EDF Expanded)
- **15 subjects** successfully preprocessed and cached as NPZ
- **1 subject** (SC4021) failed preprocessing (channel mismatch)
- **41,037 total epochs** across all cached subjects
- **N1 distribution:** 1,388 epochs (3.4%) — 4.4x increase from original 318

